

YEAR: II

DEPARTMENT: IT

MODULE NAME: Database Systems II

1. Explain the distinctions among the terms primary key, candidate key, and super key (3 marks)
2. Give an ER diagram for the COMPANY database (10 marks)
3. Give a corresponding company Relational schema (10 marks)
4. Notation Legend and supporting information should always be included alongside the answer provided in (1) & (2)

Consider the following scenario: The **COMPANY** database keeps track of a company's employees, departments, and projects:

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's name, Social Security number, address, salary, sex (gender), and birth date.
- An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department.
- We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, sex, birth date, and relationship to the employee.

4.

Below we have a college enrolment table with columns `student_id`, `subject` and `professor`.

<code>student_id</code>	<code>subject</code>	<code>professor</code>
101	Java	P.Java
101	C++	P.Cpp
102	Java	P.Java2
103	C#	P.Chash
104	Java	P.Java

In the table above:

- One student can enrol for multiple subjects. For example, student with **student_id** 101, has opted for subjects - Java & C++
- For each subject, a professor is assigned to the student.
- And, there can be multiple professors teaching one subject like we have for Java.
 - a) What do you think should be the **Primary Key**? Explain. (3 marks)
 - b) Why this table is not in BCNF? (2 marks)
 - c) Decompose this table such BCNF is satisfied (2 marks)

SECTION A: (Answer All Questions in this section)

QUESTION 1:

1. Describe ANSI/SPARC three-level architecture (external, conceptual and internal) and explain the concepts of physical and logical data independence it provides. [20 marks]
2. Explain any 3 drawbacks (of your choice) of using file based system versus database. [3 marks]
3. Explain the distinctions among the terms superkey, candidate key and primary key. [3 marks]
4. a) Explain the advantages of using Views as a way of implementing database queries. [2 marks]
b) Create a view from Loan (loan-number, amount, branch-name) that hides the amount. [1.5 marks]
c) Then find all loans in the KIGALI branch from the created view in (b). [1.5 marks]
5. What is a checkpoint? Explain its role in database backup and recovery. Discuss also how the file is a fundamental feature in any database recovery mechanism. [3 marks]
6. Explain the following threats to databases.
Loss of integrity
Loss of availability
Loss of confidentiality [3 marks]

SECTION B (Answer Any Two Questions in this section)

[15marks]

QUESTION2:

1. Consider the following scenario:

A Bus Company owns a number of busses. Each bus is allocated to a particular route, although some routes may have several busses. Each route passes through a number of towns. One or more drivers are allocated to each stage of a route, which corresponds to a journey through some or all of the towns on a route. Some of the towns have a garage where busses are kept and each of the busses are identified by the registration number and can carry different numbers of passengers, since the vehicles vary in size and can be single or double-decked. Each route is identified by a route number and information is available on the average number of passengers carried per day for each route. Drivers have an employee number, name, address, and sometimes a telephone number.

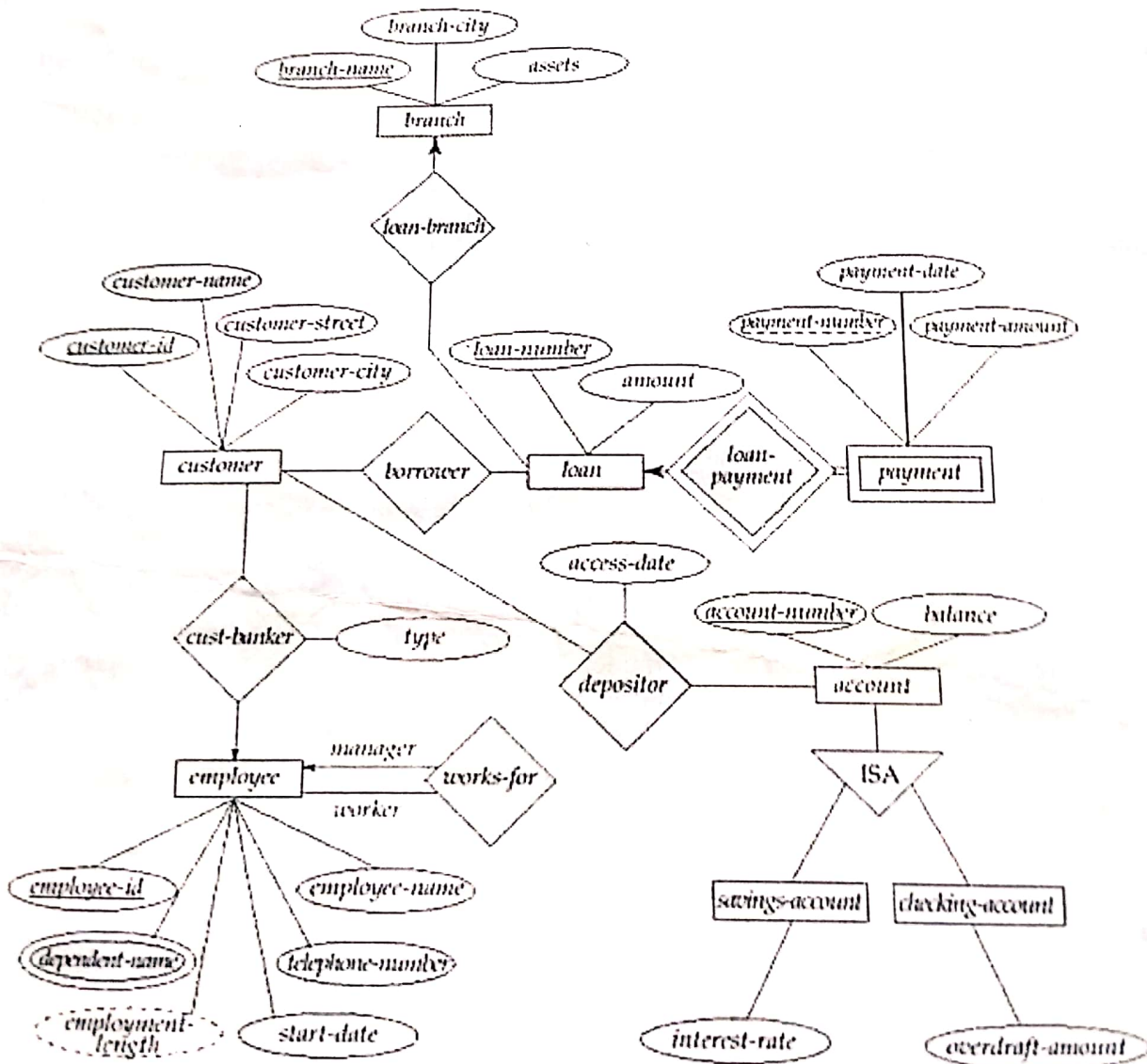
For the above scenario:

- a. List the entity types that you model (Indicate attributes and underline the primary keys) [2 marks]
- b. Show entities and explain their relationships (cardinality) [3 marks]
- c. Derive an Entity Relationship data model [10 marks]

QUESTION3:

[15marks]

2. Convert the following E-R diagram into a relational model (represent all tables, relationships, constraints and keys)



[15marks]

QUESTION4:

- a) A small banking enterprise uses a database to manage the accounts of their clients. The bank is divided into local branches, identified by unique branch codes. Accounts belong to a specific branch, and are identified by account numbers that are unique within that branch. An account can be shared by several clients, and the only other information necessary to store is the account balance.

For clients, the bank stores their social security number, first and last names, as well as a contact address and phone number.

The schema they use for their database is quite simple, and looks like this:

Clients(ssNr, firstName, lastName, address, phoneNumber)
Accounts(branchCode, branchName, accountNr, balance, client)

client → *Clients.ssNr*

This schema is not fully normalized, and thus suffers from a number of problems. It is your task to solve these by normalization of the schema.

- (i) [2 marks]

For the given domain, identify all functional dependencies that are expected to hold.

- (ii) [1 mark]

With the dependencies you have found, identify all BCNF violations in the relations of the database. For each violation, also specify what kinds of problems that could arise if it was not resolved.

- (iii) [2 marks]

Do a complete normalization of the schema, so that all relations are in BCNF. (It's the end product that's important, not the steps you take to get there.)

- b) The following relation is given: EMPLOYEE (SSN, NAME, DNO, AGE, PHONE, SALARY)
SSN is social security number, DNO is department number where employee is working,
SALARY is salary of the employee.

One wants to very quickly retrieve and analyze some salary statistics for each department and for this one needs a 'master table':

SALSTAT (DNO, SALSUM, SALAVG, SALMAX, SALMIN) which for each department maintains the max, min, average and sum of the salaries.

Show how you can maintain SALARY_DATA by using triggers so that the table SALSTAT is updated correctly whenever the table EMPLOYEE is updated.

Make sure you can handle every possible update of EMPLOYEE that influences SALSTAT, which include the following situations:

SITUATION 1: A new employee is hired

SITUATION 2: An employee resigns

SITUATION 3: The salary of an employee is changed

SITUATION 4: An employee changes a department

SITUATION 5: A department is removed.

Hint :

- separate Triggers SQL commands are needed.
 - you can combine more mentioned situations within one trigger SQL syntax.
- c) Explain ACID properties of a transaction by giving a concrete example of database transaction for each property [5 marks]

..... **GOOD LUCK!**

MODULE CODE and NAME: ITE2261 – DATABASE SYSTEMS II

SECTION A:

QUESTION 1:

[15 marks]

1. Explain main steps in database design process. [2marks]
2. Explain the distinctions among the terms primary key, candidate key, and superkey. [2 marks]
3. What is meant by a transaction and why it is an important unit of operation in a DBMS? [2 marks]
5. Define a cursor in a stored procedure [2marks]
6. Give SQL command to grant SELECT, INSERT, UPDATE, and DELETE privileges on a table called contacts to a user name john. [2marks]
7. Discuss how the log file is a fundamental feature in any database recovery mechanism. [2marks]
8. Backups of the database should be taken in order to protect data. Describe 2 features of a good backup strategy. [2marks]
9. What is the role of a Firewall in relation to Database security? [1mark]

SECTION B

QUESTION2:

[20marks]

1. A company wants to create a database to store records of its employees. For each employee, we record their personal details, the location of the branch where they work and their qualifications. Below is a sample of data for two employees.

EmployeeID: E001		Name: Fred Gordon		JobTitle: Web Developer	
BranchCode: B04			BranchAddress: Delhi		
QualificationID		Qualification		Level	
Year Obtained					
1		BSc		Undergraduate	
2012					
3		PhD		Postgraduate	
2016					

EmployeeID: E002		Name: Simon Singh	JobTitle: IT Manager
BranchCode: B01		BranchAddress: London	
QualificationID	Description	Level	YearObtained
1	BSc	Undergraduate	2009
2	MSc	Postgraduate	2010

The un-normalised table that corresponds to the above format is as follows:

Employee (EmployeeID, Name, JobTitle, BranchCode, BranchName, QualificationID, Description, Level, YearObtained)



(i) Identify the repeating group of attributes and transform the above format into tables that are in 1st Normal Form. [1 mark]

(ii) Identify any partial dependencies and transform into tables that are in 2nd Normal Form. [2 marks]

(iii) Identify any transitive dependencies and transform into tables that are in 3rd normal form [2 marks]

2.

a) Write SQL code to convert SQL query 1 into a View [3 marks]

SQL query 1:

```
SELECT cat,
AVG(price) AS AveragePrice,
SUM(price + price*0.2) AS TotalPrice
FROM titles
GROUP BY cat;
```

Create primary SQL query 1 view as
Select cat,

b) Explain the advantages of using Views as a way of implementing database queries. [2 marks]

→ Views is a mechanism used to hide

3.

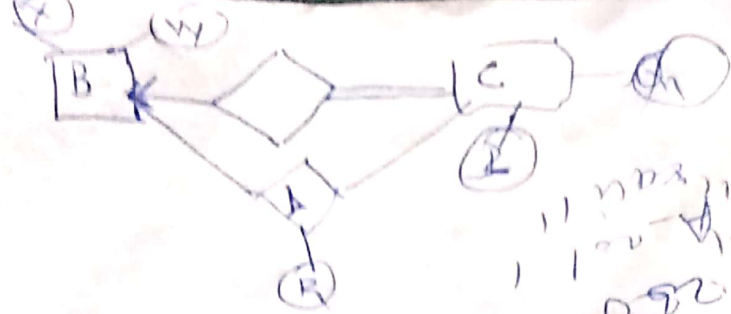
a) Suppose we decompose R into the following tables: R1 = (B, C, E) R2 = (B, C, F) R3 = (B, C, D) and R4 = (A, B, C). This decomposed schema is indeed in BCNF. Unfortunately, this decomposition is not dependency-preserving; in particular, the dependency E→F cannot be checked on a single table. A CHECK ASSERTION can be used to enforce E→F.

Complete the following SQL statement for this particular CHECK ASSERTION needed to guarantee E→F. [3 marks]

```
CREATE ASSERTION checkDep
CHECK( NOT EXISTS
( SELECT * FROM R1, R2
WHERE _____
GROUP BY _____
HAVING COUNT(_____)_____))
```

b) explain the advantages of using ASSERTION as a way of implementing database queries.

[2 marks]



Consider the following relational schema:

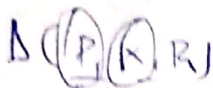
(B(X, W))

(C(P, Q, X), where X is defined as NOT NULL,

(D(P, X, R))

Give an ER diagram for the above relational schema.

[5 marks]



SECTION C

QUESTION3:

[15marks]

Consider the following scenario:

A travel company provides a selection of **Hotels** that prospective customers can reserve prior to booking a **room**. A customer can select from a range of **Accommodation Types** that each hotel offers to suit their requirements. Details of the accommodation type include the catering facilities either Self Catering (SC); Half Board (HB); Full Board (FB). The bed type either Twin bed (T); Double bed (D); Suite (S). The price of the hotel is determined by the hotel and the type of accommodation offered.

Assume that

- Each hotel is identified by a **hotel code**.
- The accommodation type is identified by a unique accommodation type code.
- Accommodation is only available during the month of **June in 2019**.

Figure below is a representative sample of data about hotels and the accommodation types offered by each hotel.

Fig: Hotel Accommodation

RESORT	HOTEL CODE	HOTEL NAME	CATERING	BED TYPE	ROOM PRICE	ACC_TYPE CODE	MONTH
Benidorm	FLB	Flamingo	SC	T	159	12	June
Palma Nova	JDM	Jardin Del Sol	SC	T	195	12	June
Benidorm	AHB	Al Hambra	FB	D	199	15	June
Santa Ponsa	HAZ	Hawaii	HB	T	308	16	June
Playa Blanca	SPZ	Sun Park	FB	S	310	18	June
Benidorm	AHB	Al Hambra	HB	S	199	17	June
Palma Nova	JDM	Jardin del Sol	FB	S	199	18	June
Palma Nova	JDM	Jardin del Sol	FB	D	169	15	June
Benidorm	SPB	Sun Park	FB	S	159	18	June

- a) Derive an Entity Relationship data model for the above scenario according to the following requirements:

[5 marks]

- Entity Types that you model are listed in bold font

- Resort varchar (20)

- Hotel varchar (10)

- Hot

- Show Relationships and participation constraints
- Resolve Many to Many relationships

- b) Derive a set of Tables/Relations from your ER model containing the sample data above.
Underline the Primary Keys in each Table. [5 marks]
- c) Extend the ER model you produced above in part a) by adding TWO further Entity Types; Customer and Reservation; to allow a Customer to reserve accommodation at a hotel possibly on different dates. [5 marks]

QUESTION 3:

[15marks]

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